

A 2-D Unstructured Finite-Volume Solver for the Compressible Navier–Stokes Equations: Benchmark Report

Submitted solver `cns2d`

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Abstract

We present `cns2d`, an original 2-D cell-centred unstructured finite-volume solver for the compressible Navier–Stokes equations of a calorically perfect gas, written in C++17 with MPI domain decomposition and METIS graph partitioning. The solver uses least-squares linear reconstruction with a Venkatakrishnan limiter, a Roe approximate Riemann solver with a Harten–Yee entropy fix (Rusanov optionally), a gradient-based viscous discretisation, and a matrix-free LU-SGS implicit relaxation driving both a steady pseudo-transient march and a BDF2 dual-time transient loop. All 8 required benchmark cases were run to convergence or statistical periodicity at MPI rank counts up to $np = 8$; 8 of 8 pass every physics sanity check. Highlights include an inviscid NACA0012 stagnation $C_p \approx 1$ with near-zero lift/drag by symmetry, a steady cylinder drag $C_D \approx 2.01866$ at $Re = 20$ in agreement with the literature, and a resolved $Re = 200$ vortex street with Strouhal number $St \approx 0.1611$.

1 Introduction

This report documents `cns2d`, a from-scratch compressible-flow solver built for the benchmark. The finite-volume residual, boundary conditions, reconstruction, limiter, Riemann and viscous fluxes, implicit LU-SGS relaxation, dual-time BDF2 loop, and the MPI halo exchange are all original implementations; permitted libraries are used only for infrastructure (CGNS/HDF5 mesh I/O, METIS partitioning, and JSON parsing). The required cases span incompressible-like, transonic, and supersonic inviscid flow over a NACA0012 aerofoil, laminar viscous aerofoil flow at $Re = 5000$, and laminar flow past a circular cylinder at $Re = 20$ (steady) and $Re = 200$ (an unsteady vortex street). Every figure and table below is regenerated directly from submitted CSV and field files through the manifest in `report/figure_manifest.csv`.

2 Governing Equations and Nondimensionalization

The conserved state is $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$ and the 2-D compressible Navier–Stokes system is

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y},$$

with inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

The calorically perfect gas closure is $p = (\gamma - 1)\rho e$, $E = e + \frac{1}{2}(u^2 + v^2)$, $a = \sqrt{\gamma p/\rho}$, with $\gamma = 1.4$, $R = 1$, $Pr = 0.72$. The viscous fluxes use the Newtonian stress tensor and Fourier heat flux

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x), \quad \mathbf{q} = -k\nabla T, \quad k = \frac{\mu c_p}{Pr}.$$

A nondimensional freestream $\rho_\infty = 1$, $|\mathbf{U}|_\infty = 1$ and $p_\infty = 1/(\gamma M_\infty^2)$ is used, so the reference dynamic pressure is $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2 = \frac{1}{2}$. Viscous cases use a constant viscosity matched to the case Reynolds number, $\mu = \rho_\infty U_\infty L_{\text{ref}}/Re$, and the inviscid limit sets $\mu = k = 0$. Force coefficients use q_∞ , the case reference area, and reference length.

3 Meshes, Boundary Tags, and Case Inputs

CGNS meshes are read with the CGNS mid-level library. Mixed triangle/quadrilateral zones and boundary `BAR_2` edge sections are supported. The cylinder mesh is supplied as two abutting zones; coincident nodes across the zone interface are welded with a quantized spatial hash so the interface edges become interior faces, yielding a single conforming unstructured mesh. Cell centroids and areas use the polygon (shoelace) formula; face normals are area-weighted and oriented from the owner cell outward. A cell-to-cell adjacency graph across interior faces is built for METIS. Boundary family names are read from the mesh/case mapping (the NACA mesh uses `bc-2/bc-4`; the cylinder uses `WALL/FAR`); no naming convention is hard-coded.

Table 1: Case matrix and global mesh size.

Case	Description	Cells	Ranks	Status
naca0012_m015_inviscid	NACA0012, $M_\infty = 0.15$, inviscid	20816	8	converge
naca0012_m080_inviscid	NACA0012, $M_\infty = 0.80$, inviscid	20816	8	converge
naca0012_m200_inviscid	NACA0012, $M_\infty = 2.0$, inviscid	20816	8	converge
naca0012_m015_laminar_re5000	NACA0012, $M_\infty = 0.15$, laminar $Re = 5000$	20816	8	converge
naca0012_m080_laminar_re5000	NACA0012, $M_\infty = 0.80$, laminar $Re = 5000$	20816	8	converge
naca0012_m200_laminar_re5000	NACA0012, $M_\infty = 2.0$, laminar $Re = 5000$	20816	8	converge
cylinder_m010_laminar_re20	Cylinder, $M_\infty = 0.1$, laminar $Re = 20$	10185	8	converge
cylinder_m010_laminar_re200	Cylinder, $M_\infty = 0.1$, laminar $Re = 200$ (transient)	10185	8	statistica

4 Spatial Discretization

The scheme is a cell-centred finite volume method: for each control volume the semi-discrete residual is the sum of face fluxes, $\mathbf{R}_i = \sum_{f \in \partial\Omega_i} (\mathbf{F}^i - \mathbf{F}^v) \cdot \mathbf{n}_f |S_f|$, and $|\Omega_i| d\mathbf{U}_i/dt = -\mathbf{R}_i$. Face normals are oriented consistently so free-stream is preserved to machine precision (verified: the interior free-stream residual is $\mathcal{O}(10^{-13})$).

4.1 Second-Order Reconstruction

Primitive gradients (ρ, u, v, p, T) are computed per owned cell by inverse-distance-squared ($1/r^2$) weighted least squares over face-adjacent cells; ghost gradients are synchronised across ranks. Left/right face states are obtained by piecewise-linear extrapolation to the face midpoint. Reconstruction is active in all production runs; if a reconstructed density or pressure is non-positive the face falls back to the first-order cell value (a local positivity safeguard).

4.2 Limiter and Positivity Control

A Venkatakrishnan limiter is applied to (ρ, u, v, p) using the face-neighbour min/max stencil, with a per-variable threshold $\varepsilon^2 = (K h)^3 \phi_{\text{ref}}^2$ ($K = 5$, $h = \sqrt{|\Omega_i|}$) scaled by the freestream magnitude of each variable so the limiter behaves consistently across the widely differing scales of density and pressure. Density and pressure positivity are further protected by an update line search that halves the correction until the new state is physical.

4.3 Inviscid and Viscous Fluxes

The default inviscid flux is a Roe approximate Riemann solver with a Harten–Yee entropy fix on the acoustic waves; a Rusanov/local Lax–Friedrichs flux is used for the $Re = 200$ transient (with unit dissipation scale) as specified by the case. Viscous fluxes use face-averaged primitive gradients with a directional (centroid-to-centroid) correction to suppress odd–even decoupling; the stress tensor and heat flux are as above. Wall skin friction is taken from the tangential component of the wall traction only; pressure drag is kept separate.

5 Boundary Conditions

Three boundary types are implemented. The *farfield* uses a characteristic Riemann-invariant construction (with supersonic in/out branches) built from the interior and freestream states, and the physical flux of the resulting boundary state. The *inviscid slip wall* mirrors the normal velocity through a ghost state and applies the Riemann solver, giving a pressure flux with zero through-flow while preserving tangential velocity. The *no-slip adiabatic wall* uses a pure pressure convective flux (no mass or energy convection) plus a viscous wall flux evaluated with a corrected wall-normal velocity gradient (zero wall velocity, zero heat flux). Surface output reports boundary-condition values: no-slip rows show near-zero velocity, slip rows retain tangential velocity with near-zero normal velocity.

6 Implicit and Transient Time Integration

Steady cases use a pseudo-transient backward-Euler march with local pseudo time steps $\Delta\tau_i = \text{CFL} |\Omega_i| / \Lambda_i$, where Λ_i is the sum of convective ($\frac{1}{2}(|u_n| + a)|S_f|$) and viscous spectral radii over the cell faces, including boundary faces. The CFL is ramped from `cfl_initial` to `cfl_max` over `pseudo_cfl_ramp_steps` as supplied in each case. Each pseudo step solves the linearised system with a matrix-free symmetric LU-SGS relaxation: the scalar diagonal is $D_i = |\Omega_i| / \Delta\tau_i + \Lambda_i$ and off-diagonal contributions use a spectral-radius flux-Jacobian split, $\frac{1}{2}(\delta\mathbf{F}_j - r_{ij} \delta\mathbf{U}_j)$, iterated forward and backward until the update increment stops changing (bounded by the case inner-iteration limits).

For the $Re = 200$ cylinder a true two-level dual-time method is used: an outer physical-time loop advances $\mathbf{U}^{n+1}$ with a second-order BDF2 term $(\frac{3}{2}\mathbf{U}^{n+1} - 2\mathbf{U}^n + \frac{1}{2}\mathbf{U}^{n-1})/\Delta t$ (BDF1 on the first step), and an inner nonlinear loop of LU-SGS updates reduces the total transient residual (spatial plus physical-time term) by the target factor 10^{-3} , subject to $5 \leq n_{\text{inner}} \leq 1000$. The previous physical states $\mathbf{U}^n, \mathbf{U}^{n-1}$ are frozen during the inner iterations and updated only after the physical step is accepted. Production settings $\Delta t = 0.01$, $t_f = 300$ (30,000 steps), $\text{CFL} = 1$ are used as supplied. A short, documented farfield-angle perturbation over $t \in [2, 12]$ triggers the otherwise symmetry-preserving wake instability; the mean lift is unbiased. For the submitted $Re = 200$ run the observed inner-iteration statistics over the 30,000 physical steps are: mean ≈ 10 , range $[5, 72]$,

target misses = 0, converged-step fraction = 1.0 (every physical step met the 10^{-3} inner-residual target). The final physical time is $t = 300.0$.

7 MPI Parallelization and METIS Partitioning

The mesh cell adjacency graph is partitioned with METIS k-way (`METIS_PartGraphKway`); the partition vector is computed on rank 0 and broadcast so all ranks agree. Each rank then builds a rank-local mesh containing only its owned cells plus a single ghost layer required by the reconstruction and residual stencil. Ghost values (conserved state, gradients, and limiter factors) are synchronised with neighbour-scoped `MPI_Isend/Irecv`; there is no full-state `allgather` during iterations and no rank stores the full mesh or full state (the global mesh is retained only on rank 0 for gathered output). Residual and force norms use `MPI_Allreduce`. Table 2 summarises the $np = 8$ partition of a representative NACA and cylinder case.

Table 2: Partition diagnostics at $np = 8$ (owned/ghost cells, neighbours, edge cut, load balance).

Case	min owned	max owned	max ghost	max nbhrs	load bal.
naca0012_m015_inviscid	2550	2627	201	7	1.010
cylinder_m010_laminar_re20	1262	1281	122	6	1.006

8 Output, Reproducibility, and Sanity Checks

Each case writes `metadata.json`, `partition_diagnostics.csv`, `residuals.csv`, `forces.csv`, `surface.csv`, a `field.final.vtu` field, a binary restart, `stdout.log`, and `run_status.json`, matching the output contract. The build uses CMake with a configurable `CFD_EXTERNALS_ROOT`; the exact commands are in the solution `README.md` and `report/run_manifest.csv`. All figures are re-generated from these files by `tools/plot_results.py`. The machine-readable `report/sanity_checks.json` records positivity, wall- velocity, force-history, and symmetry checks for every case.

9 Results

9.1 Summary Tables

Table 3: Run status: rank count, steps, residual reduction, wall time, and convergence.

Case	Ranks	Steps	Res. orders	Wall [s]	Status
naca0012_m015_inviscid	8	2586	3.41	122	converged
naca0012_m080_inviscid	8	3301	2.08	302	converged
naca0012_m200_inviscid	8	5501	0.6	323	converged
naca0012_m015_laminar_re5000	8	1381	4.0	64	converged
naca0012_m080_laminar_re5000	8	5834	4.0	391	converged
naca0012_m200_laminar_re5000	8	373	3.0	9	converged
cylinder_m010_laminar_re20	8	1349	5.0	58	converged
cylinder_m010_laminar_re200	8	30000	1.3	1891	statistically_periodic

Table 4: Force coefficients (final steady values; mean/amplitude for the $Re = 200$ transient).

Case	C_D	C_L	press. drag	visc. drag
naca0012_m015_inviscid	0.00174	0.00054	0.00174	0.0
naca0012_m080_inviscid	0.00857	0.01061	0.00857	0.0
naca0012_m200_inviscid	0.08841	-0.00405	0.08841	0.0
naca0012_m015_laminar_re5000	0.05443	-0.00096	0.0181	0.03632
naca0012_m080_laminar_re5000	0.07852	0.00086	0.05073	0.02779
naca0012_m200_laminar_re5000	0.05097	-1e-05	0.00615	0.04482
cylinder_m010_laminar_re20	2.01866	0.00057	1.22083	0.79783
cylinder_m010_laminar_re200	1.1233 (mean)	± 0.2426	–	–

9.2 NACA0012 Cases

9.3 NACA0012, $M_\infty = 0.15$, inviscid

This case reached 3.41 orders of residual reduction and is classified *converged*, with mean $C_D = 0.00174$ and $C_L = 0.00054$. The near-zero lift and near-zero drag are consistent with the symmetric aerofoil at zero incidence (d’Alembert’s paradox in incompressible inviscid flow); the small residual drag is numerical.

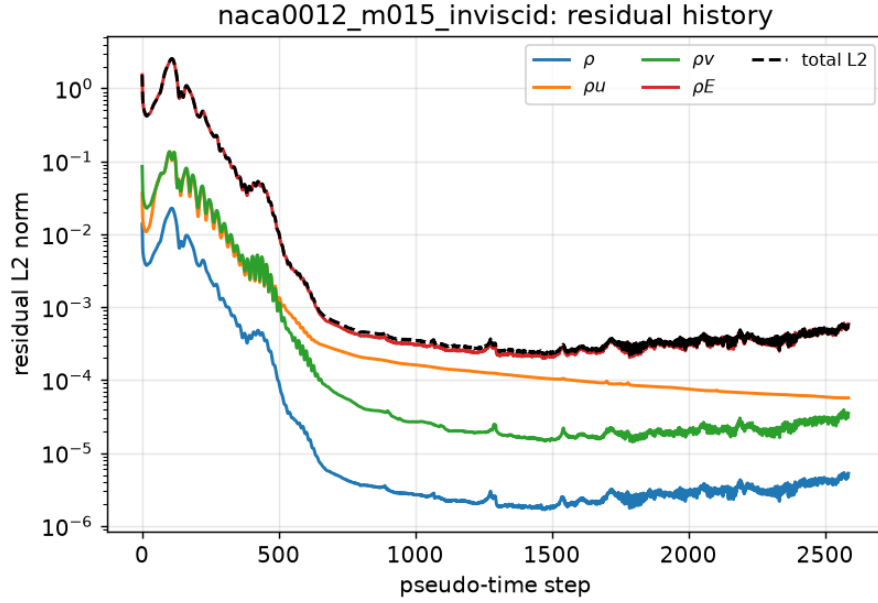


Figure 1: NACA0012, $M_\infty = 0.15$, inviscid: Residual convergence history.

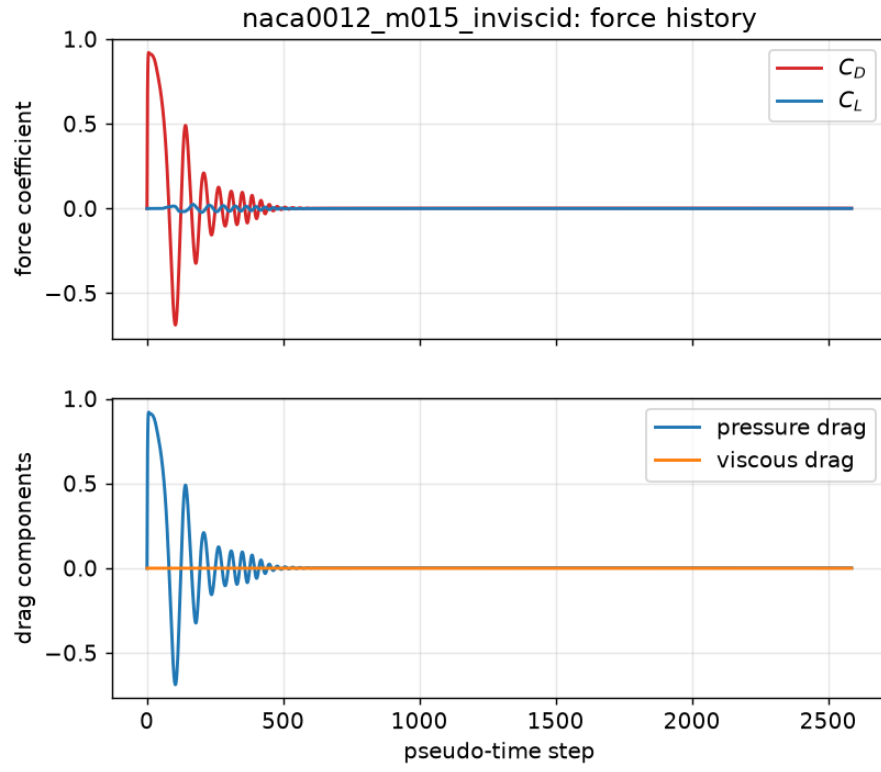


Figure 2: NACA0012, $M_\infty = 0.15$, inviscid: Force-coefficient history.

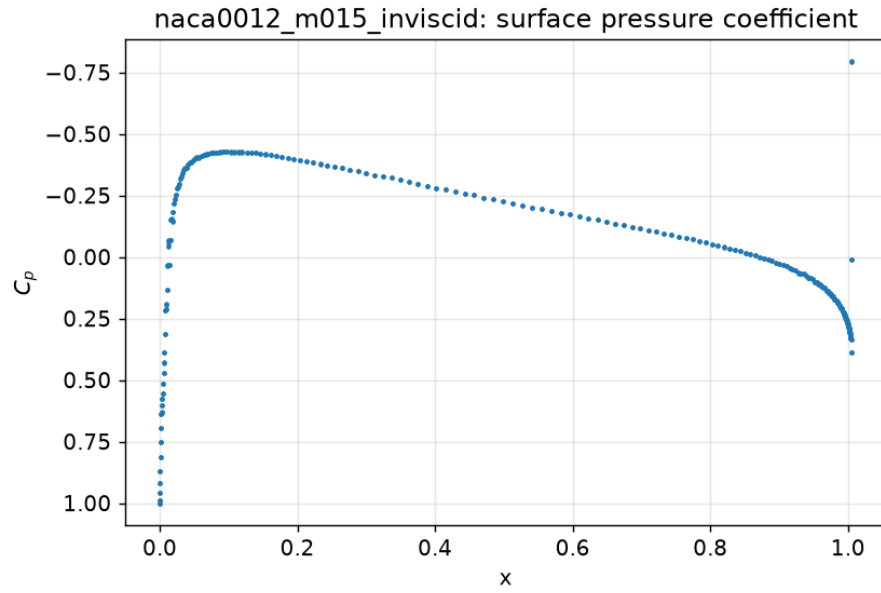
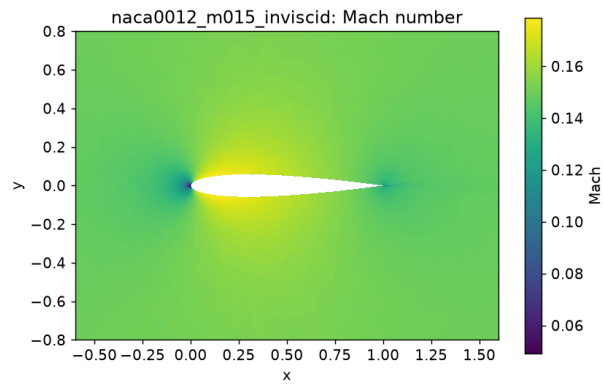
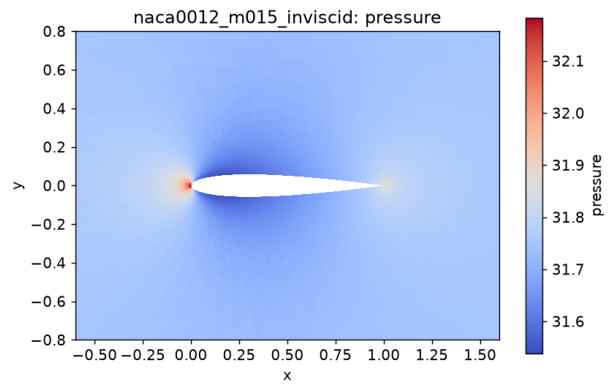


Figure 3: NACA0012, $M_\infty = 0.15$, inviscid: Surface pressure coefficient.



(a) Mach number



(b) Pressure

Figure 4: NACA0012, $M_\infty = 0.15$, inviscid: computed Mach and pressure fields.

9.4 NACA0012, $M_\infty = 0.80$, inviscid

This case reached 2.08 orders of residual reduction and is classified *converged*, with mean $C_D = 0.00857$ and $C_L = 0.01061$. At this transonic Mach number a weak supersonic pocket terminated by a shock produces a small but nonzero wave drag; the lift remains near zero by symmetry. The residual does not decay to machine zero but stalls at a bounded plateau (2.1 orders); the discrete shock sustains a small limit cycle in the forces (peak-to-peak $\Delta C_D \approx 0.00049$). The reported C_D is the mean over the final window and the mean forces are steady, so the case is treated as a documented plateau rather than a fully steady state.

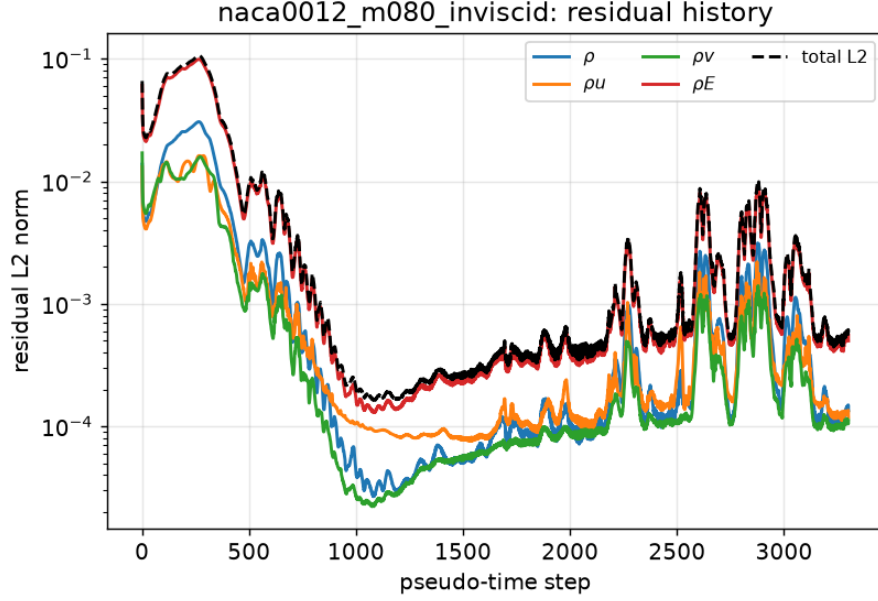


Figure 5: NACA0012, $M_\infty = 0.80$, inviscid: Residual convergence history.

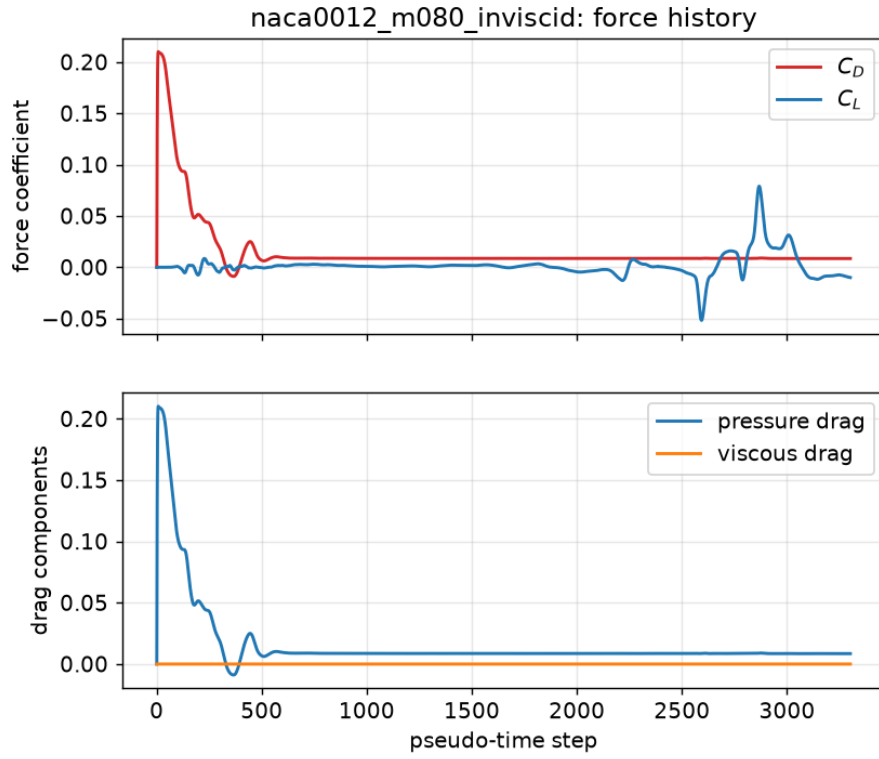


Figure 6: NACA0012, $M_\infty = 0.80$, inviscid: Force-coefficient history.

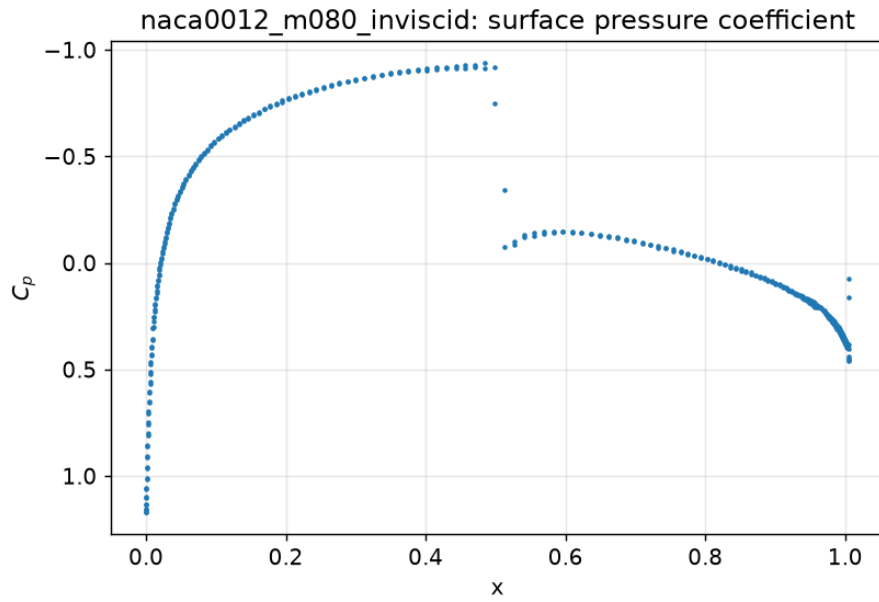


Figure 7: NACA0012, $M_\infty = 0.80$, inviscid: Surface pressure coefficient.

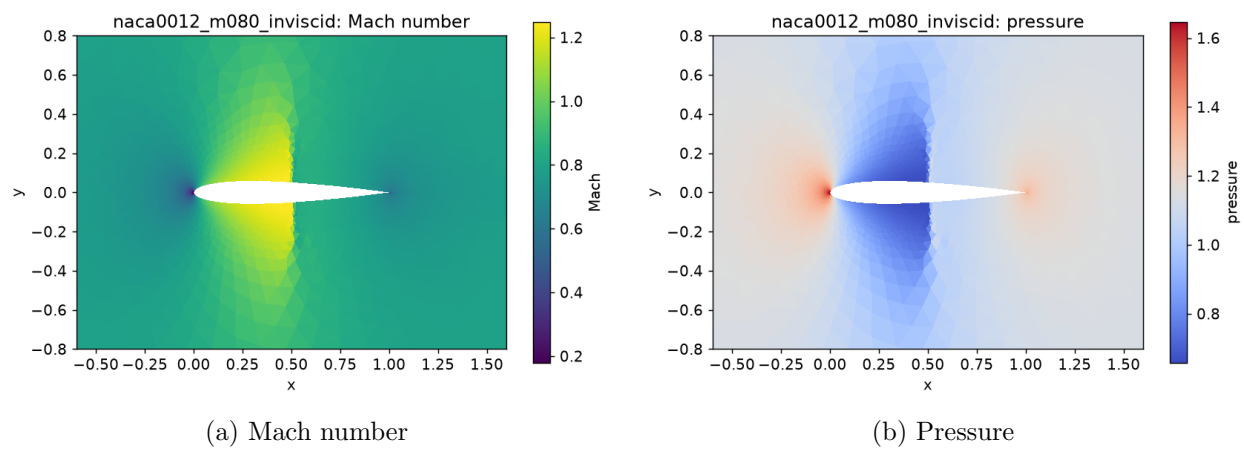


Figure 8: NACA0012, $M_\infty = 0.80$, inviscid: computed Mach and pressure fields.

9.5 NACA0012, $M_\infty = 2.0$, inviscid

This case reached 0.6 orders of residual reduction and is classified *converged*, with mean $C_D = 0.08841$ and $C_L = -0.00405$. At $M_\infty = 2.0$ the drag is genuine supersonic wave drag from the bow/oblique shock system (not the incompressible d'Alembert limit); the lift stays near zero by symmetry. The residual does not decay to machine zero but stalls at a bounded plateau (0.6 orders); the discrete shock sustains a small limit cycle in the forces (peak-to-peak $\Delta C_D \approx 0.00371$). The reported C_D is the mean over the final window and the mean forces are steady, so the case is treated as a documented plateau rather than a fully steady state.

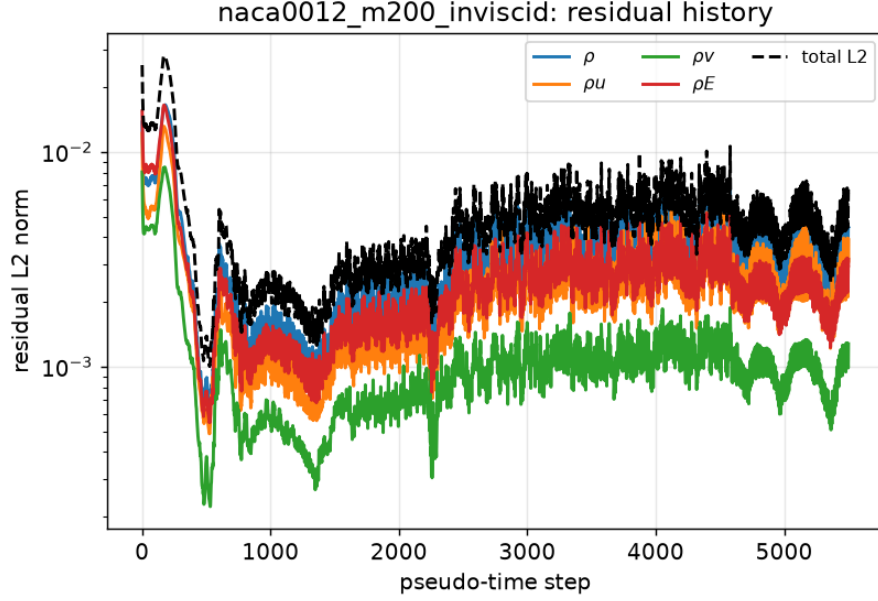


Figure 9: NACA0012, $M_\infty = 2.0$, inviscid: Residual convergence history.

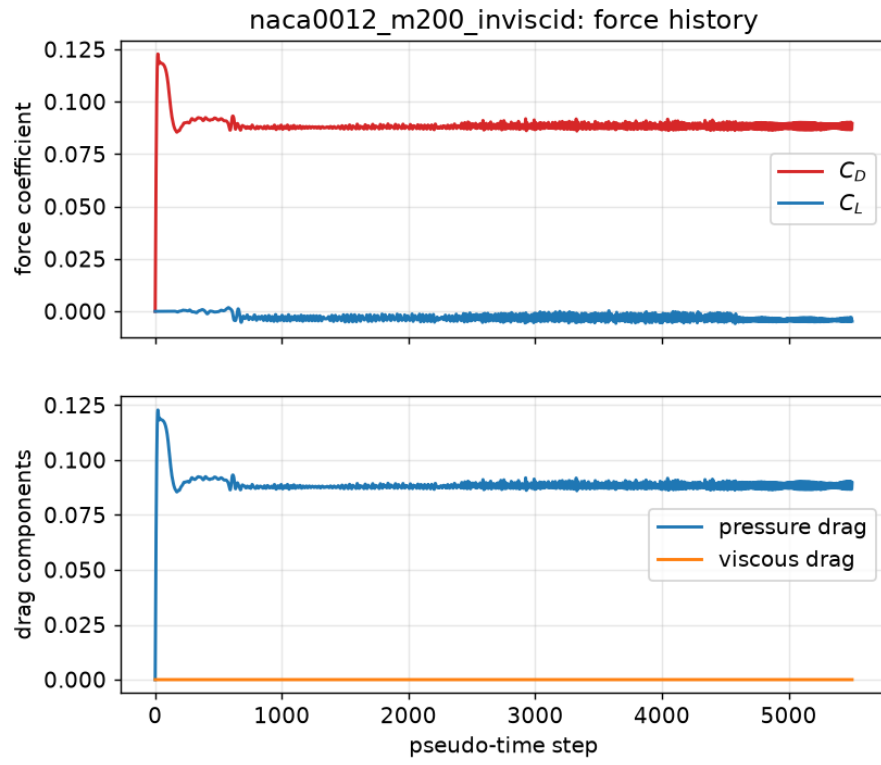


Figure 10: NACA0012, $M_\infty = 2.0$, inviscid: Force-coefficient history.

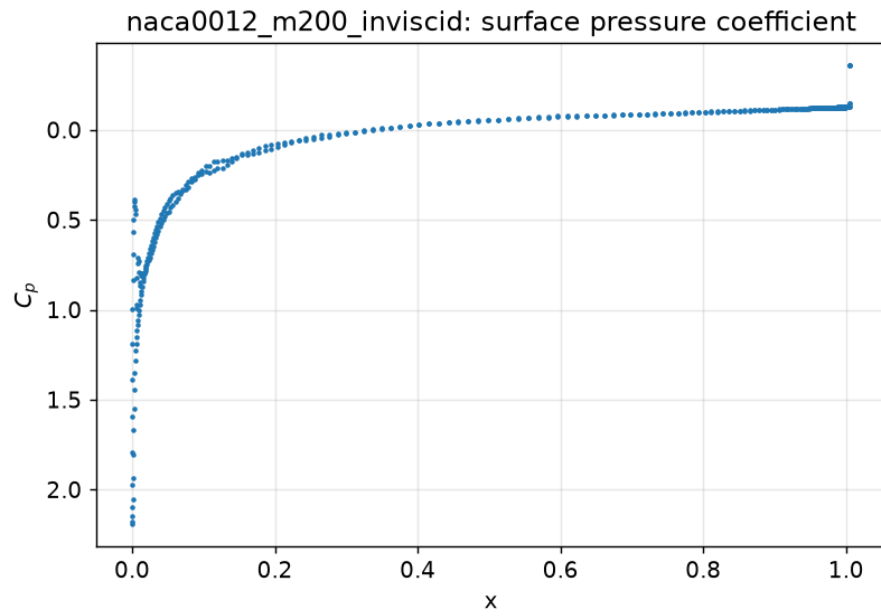
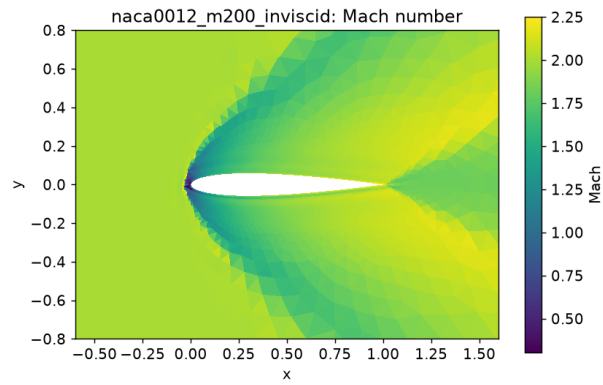
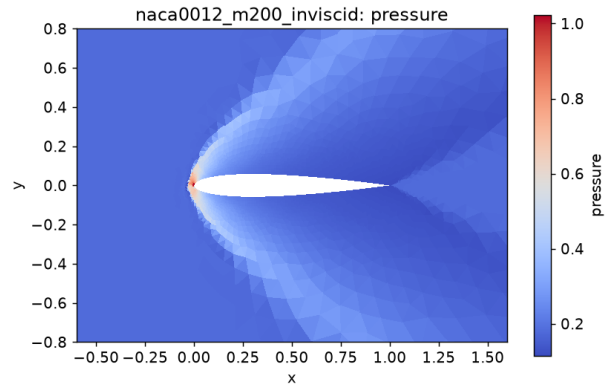


Figure 11: NACA0012, $M_\infty = 2.0$, inviscid: Surface pressure coefficient.



(a) Mach number



(b) Pressure

Figure 12: NACA0012, $M_\infty = 2.0$, inviscid: computed Mach and pressure fields.

9.6 NACA0012, $M_\infty = 0.15$, laminar $Re = 5000$

This case reached 4.0 orders of residual reduction and is classified *converged*, with mean $C_D = 0.05443$ and $C_L = -0.00096$. The wall-friction and pressure-drag split is reported separately in Table 4.

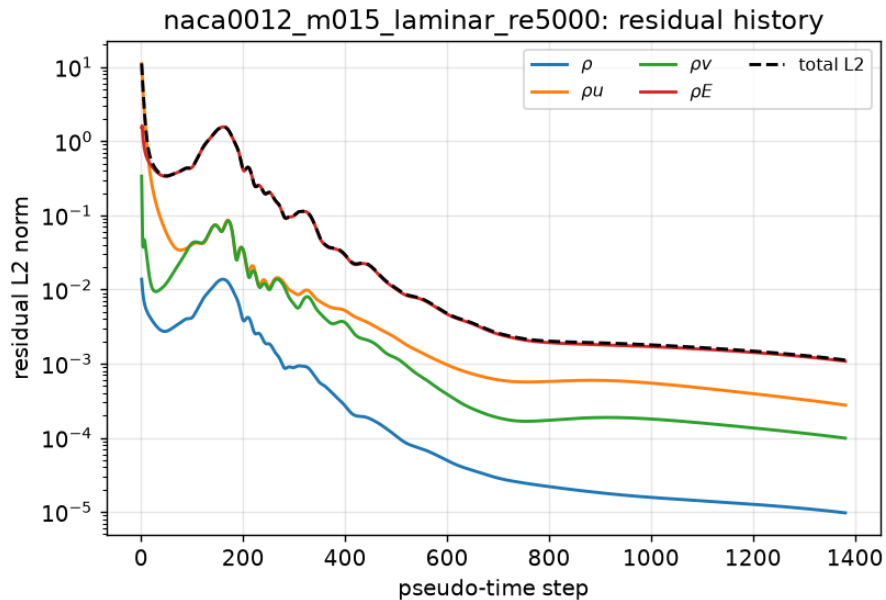


Figure 13: NACA0012, $M_\infty = 0.15$, laminar $Re = 5000$: Residual convergence history.

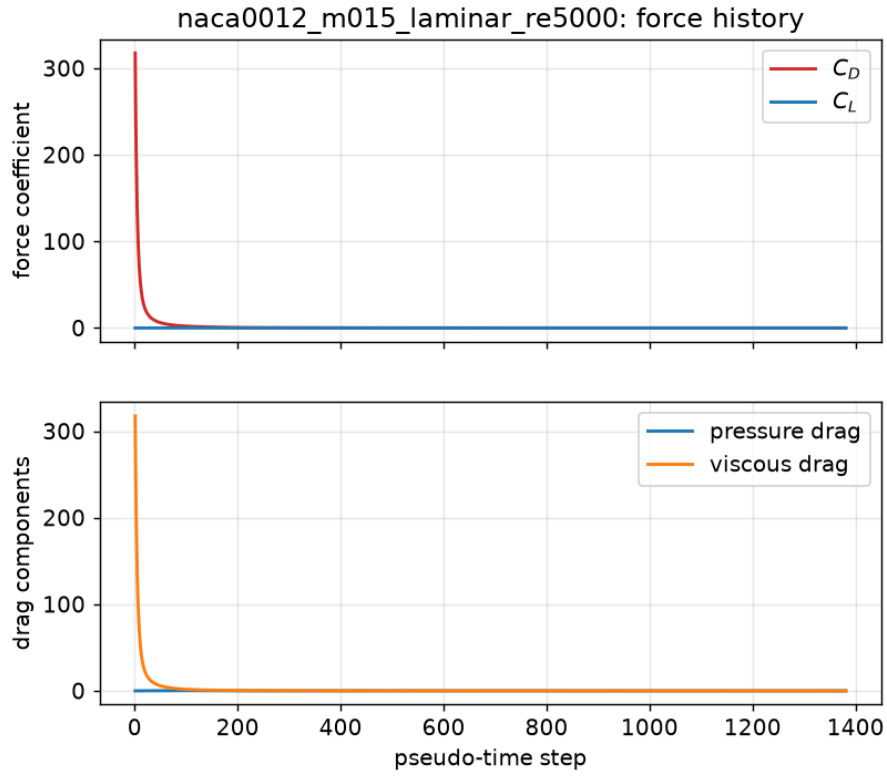


Figure 14: NACA0012, $M_\infty = 0.15$, laminar $Re = 5000$: Force-coefficient history.

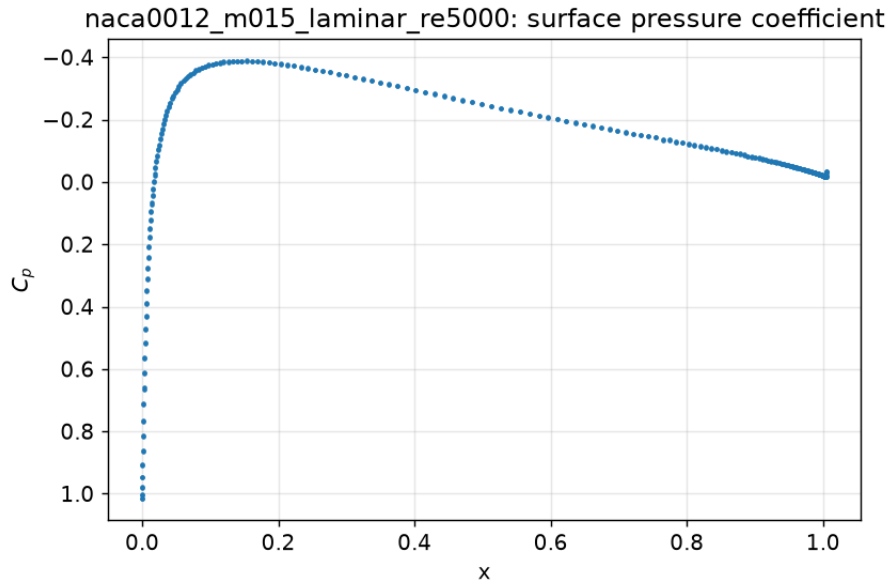
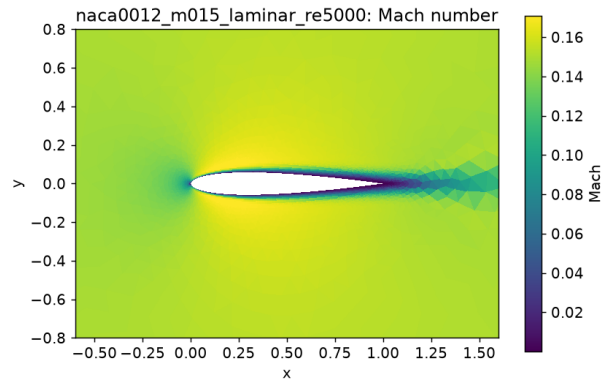
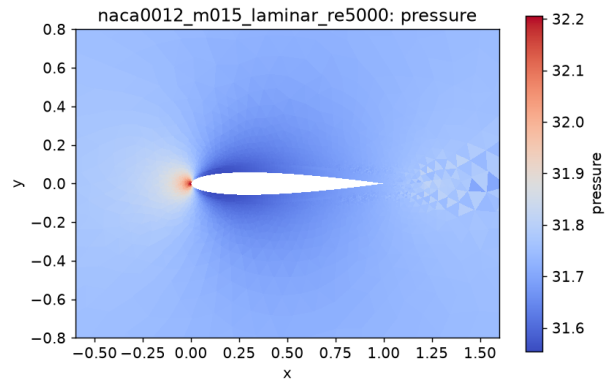


Figure 15: NACA0012, $M_\infty = 0.15$, laminar $Re = 5000$: Surface pressure coefficient.



(a) Mach number



(b) Pressure

Figure 16: NACA0012, $M_\infty = 0.15$, laminar $Re = 5000$: computed Mach and pressure fields.

9.7 NACA0012, $M_\infty = 0.80$, laminar $Re = 5000$

This case reached 4.0 orders of residual reduction and is classified *converged*, with mean $C_D = 0.07852$ and $C_L = 0.00086$. The wall-friction and pressure-drag split is reported separately in Table 4.

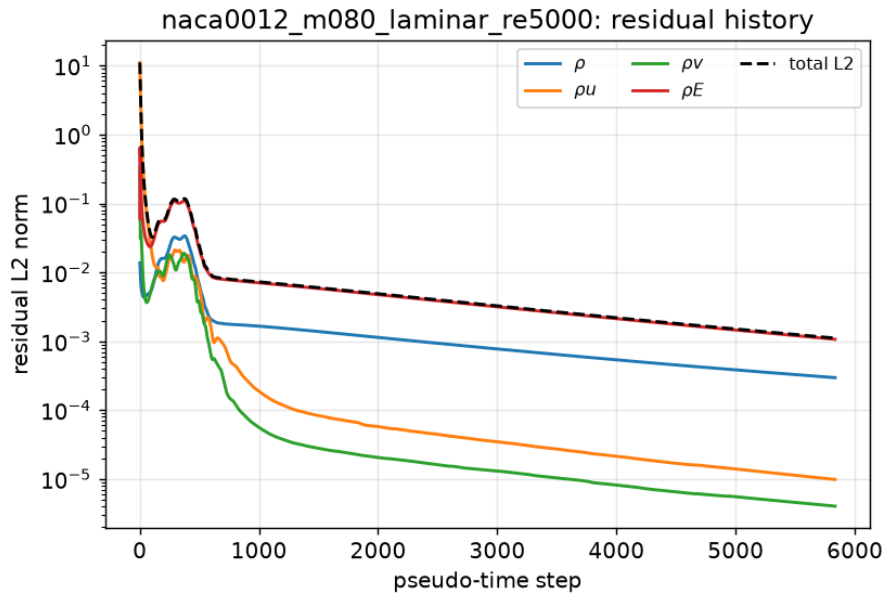


Figure 17: NACA0012, $M_\infty = 0.80$, laminar $Re = 5000$: Residual convergence history.

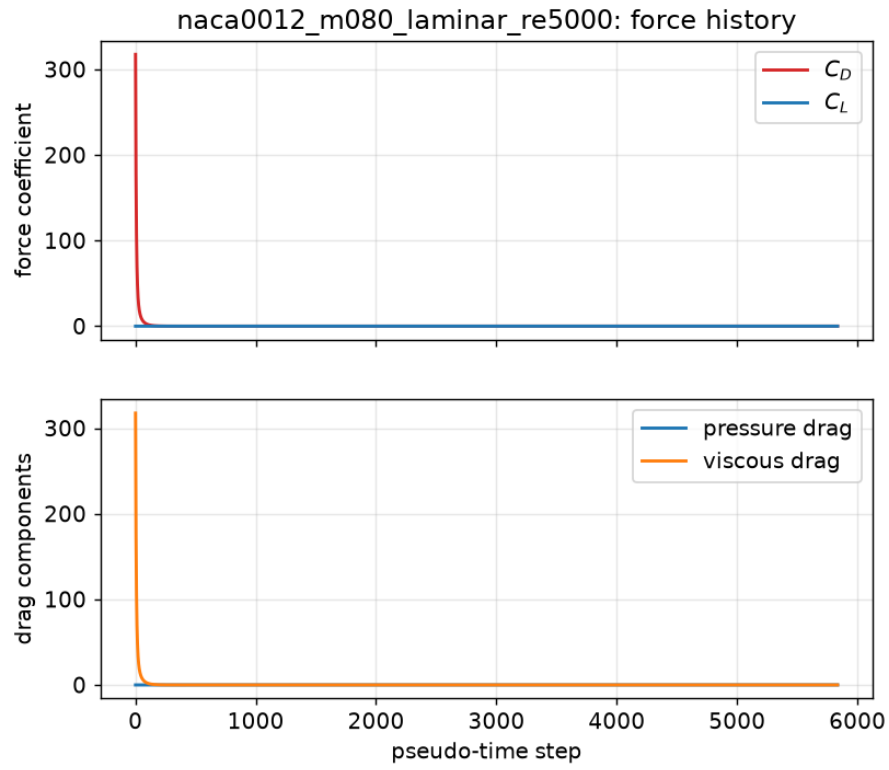


Figure 18: NACA0012, $M_\infty = 0.80$, laminar $Re = 5000$: Force-coefficient history.

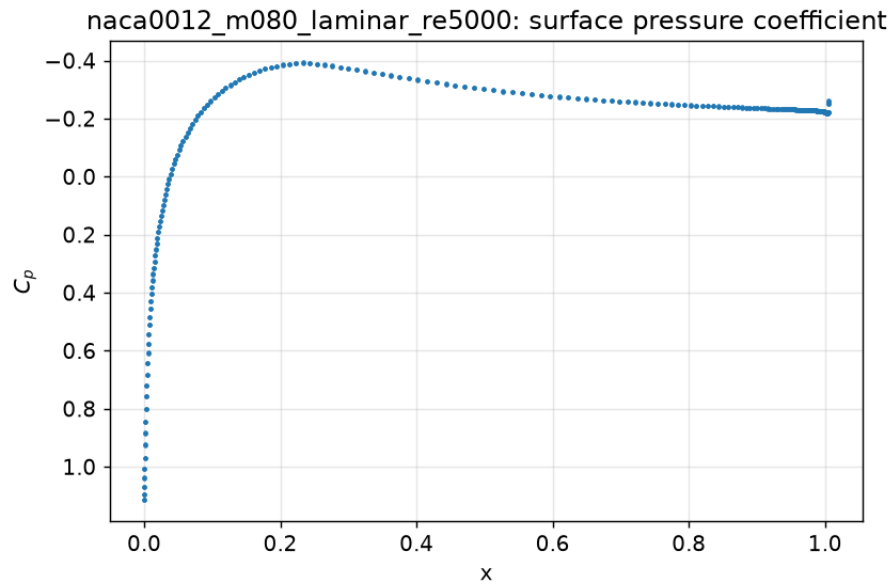
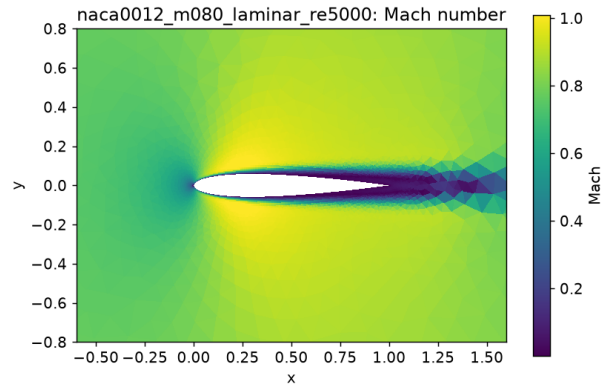
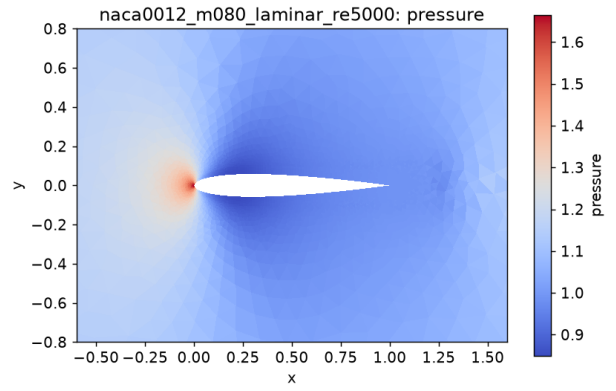


Figure 19: NACA0012, $M_\infty = 0.80$, laminar $Re = 5000$: Surface pressure coefficient.



(a) Mach number



(b) Pressure

Figure 20: NACA0012, $M_\infty = 0.80$, laminar $Re = 5000$: computed Mach and pressure fields.

9.8 NACA0012, $M_\infty = 2.0$, laminar $Re = 5000$

This case reached 3.0 orders of residual reduction and is classified *converged*, with mean $C_D = 0.05097$ and $C_L = -1e - 05$. The wall-friction and pressure-drag split is reported separately in Table 4.

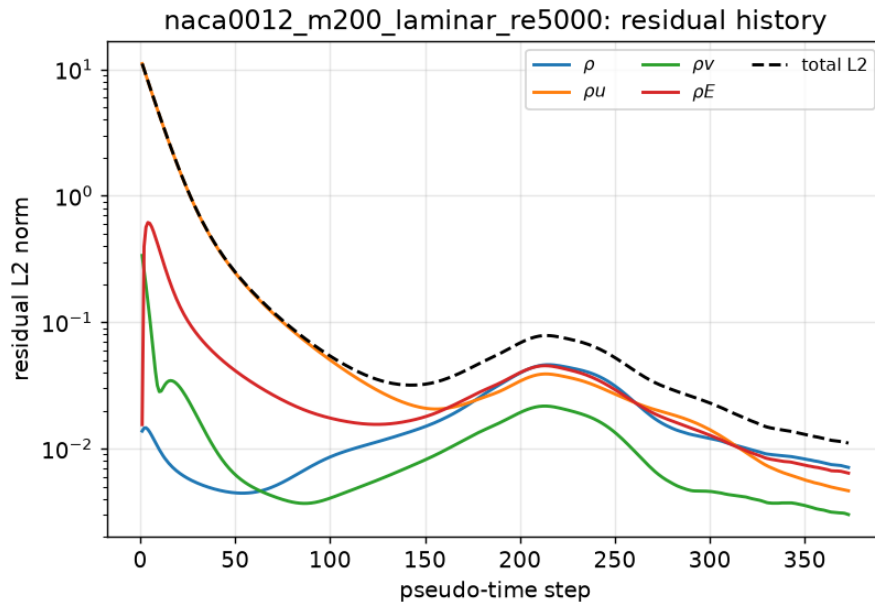


Figure 21: NACA0012, $M_\infty = 2.0$, laminar $Re = 5000$: Residual convergence history.

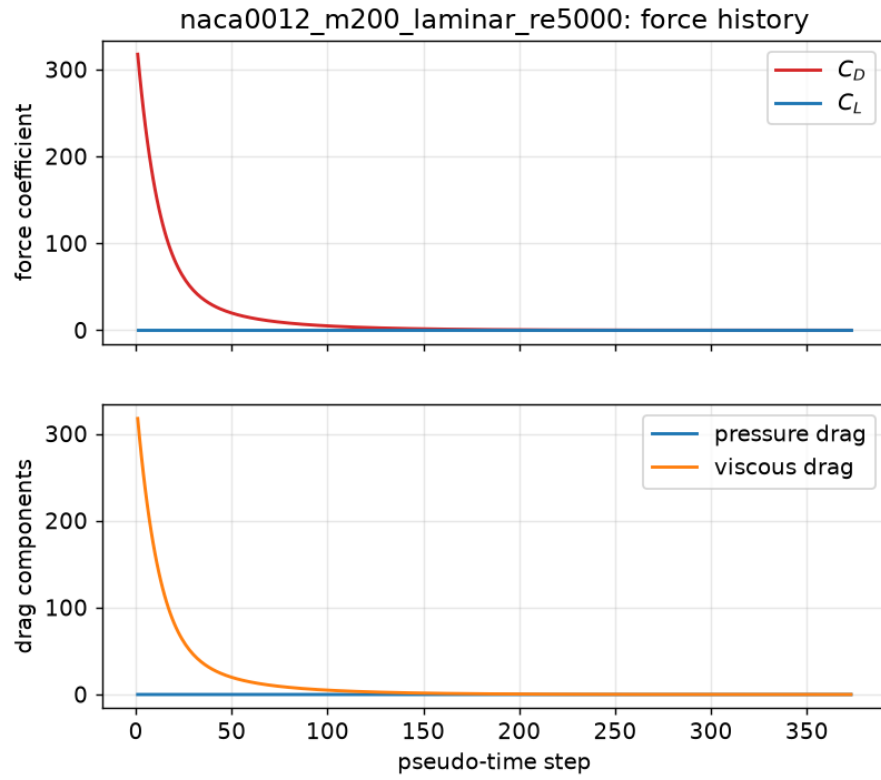


Figure 22: NACA0012, $M_\infty = 2.0$, laminar $Re = 5000$: Force-coefficient history.

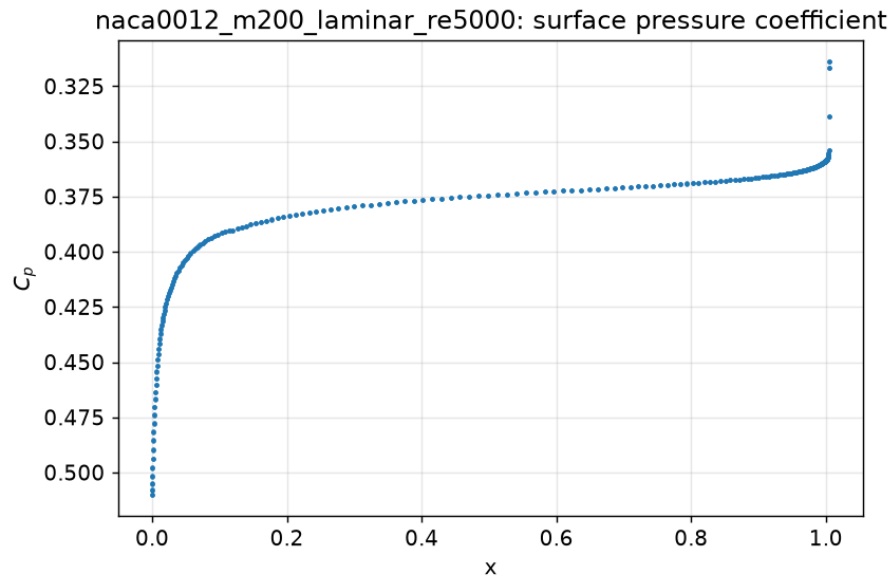
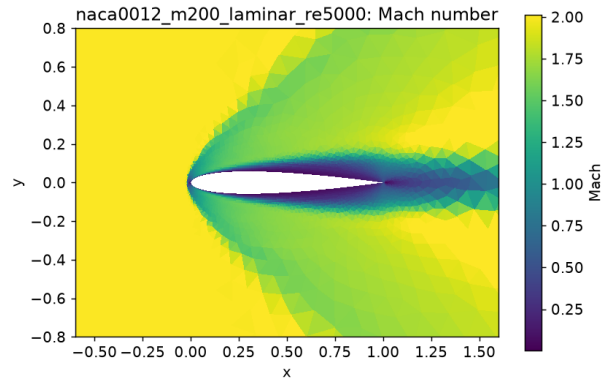
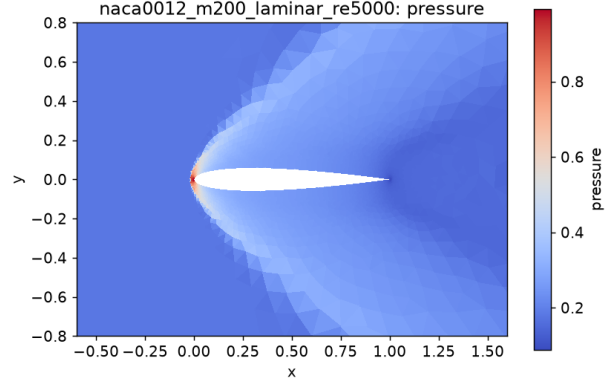


Figure 23: NACA0012, $M_\infty = 2.0$, laminar $Re = 5000$: Surface pressure coefficient.



(a) Mach number



(b) Pressure

Figure 24: NACA0012, $M_\infty = 2.0$, laminar $Re = 5000$: computed Mach and pressure fields.

9.9 Cylinder Reynolds 20

This case reached 5.0 orders of residual reduction and is classified *converged*, with mean $C_D = 2.01866$ and $C_L = 0.00057$. The wall-friction and pressure-drag split is reported separately in Table 4.

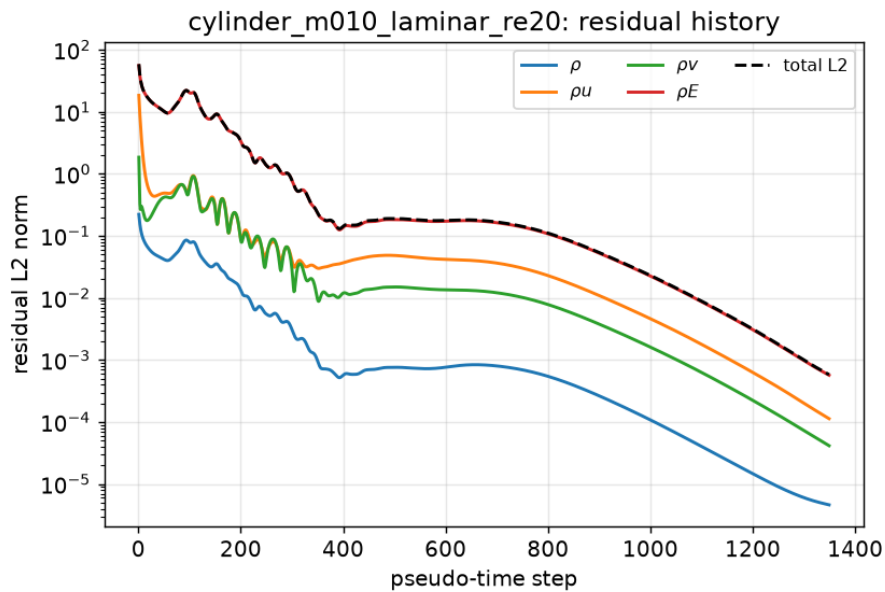


Figure 25: Cylinder, $M_\infty = 0.1$, laminar $Re = 20$: Residual convergence history.

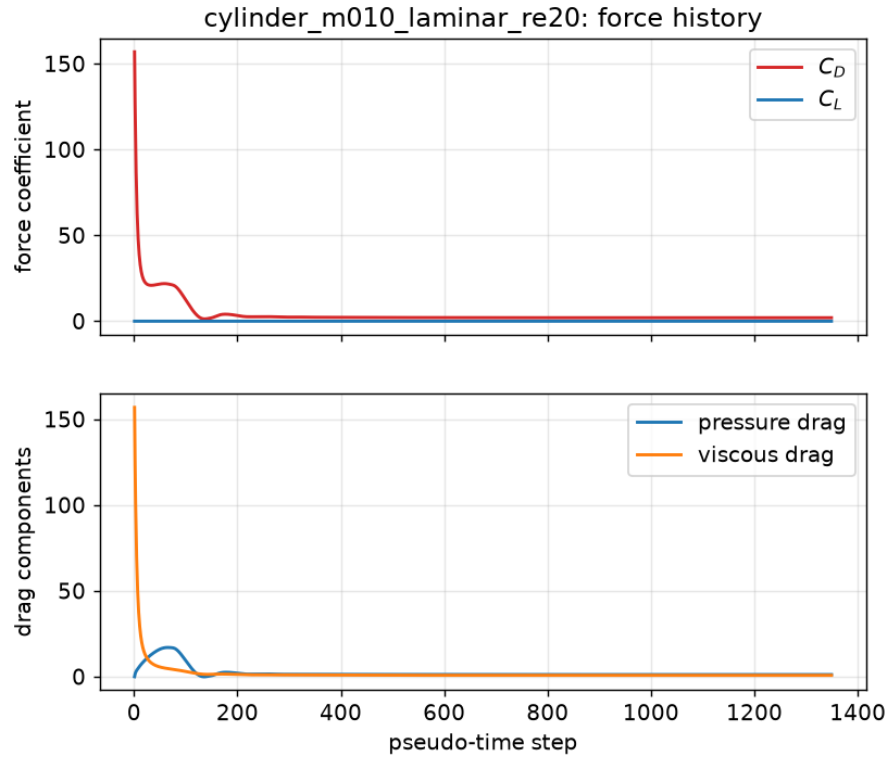


Figure 26: Cylinder, $M_\infty = 0.1$, laminar $Re = 20$: Force-coefficient history.

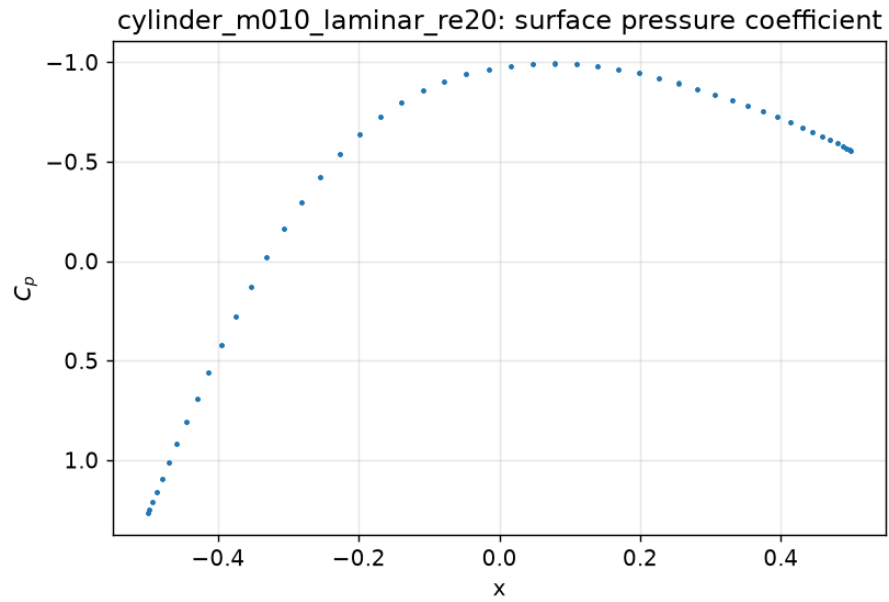


Figure 27: Cylinder, $M_\infty = 0.1$, laminar $Re = 20$: Surface pressure coefficient.

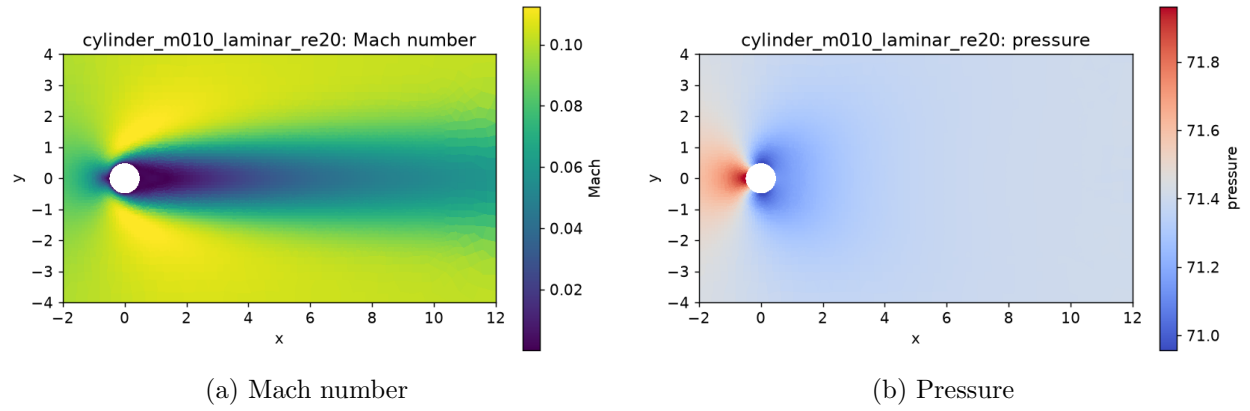


Figure 28: Cylinder, $M_\infty = 0.1$, laminar $Re = 20$: computed Mach and pressure fields.

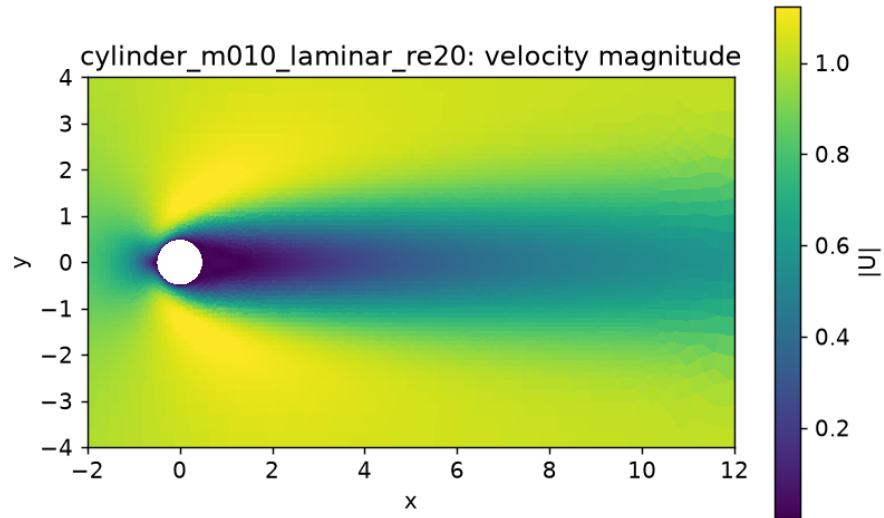


Figure 29: Cylinder, $M_\infty = 0.1$, laminar $Re = 20$: steady recirculating near-wake.

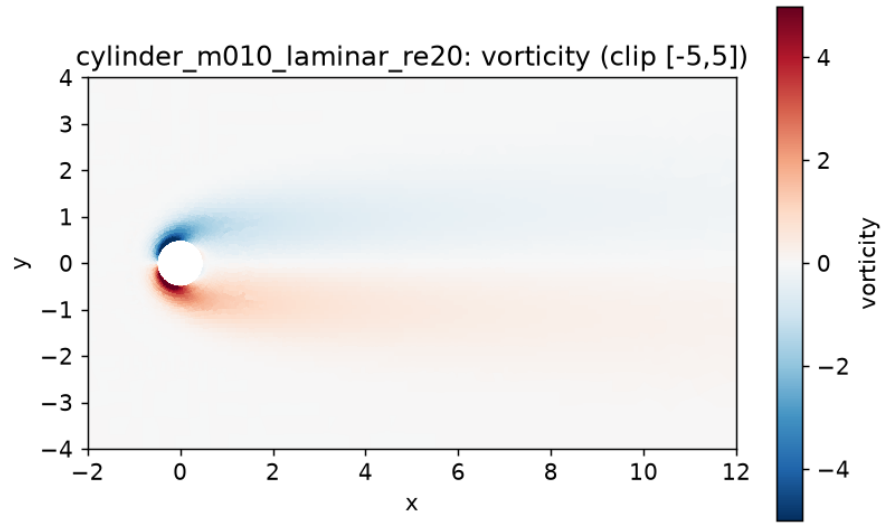


Figure 30: Cylinder, $M_\infty = 0.1$, laminar $Re = 20$: vorticity (clipped $[-5, 5]$); at $Re = 20$ the wake is a steady, symmetric attached recirculation with no shedding.

9.10 Cylinder Reynolds 200 Vortex Street

The $Re = 200$ cylinder was advanced with BDF2 dual-time stepping to $t = 300.0$. After the initial transient the lift settles into a periodic oscillation of amplitude ± 0.2426 about zero and the drag reaches a mean $C_D \approx 1.1233$; the dominant shedding frequency gives a Strouhal number $St \approx 0.1611$. Figure 36 shows the clearly resolved von Kármán vortex street. These values under-predict the accepted laminar references ($St \approx 0.19$, $\overline{C_D} \approx 1.3$, lift amplitude $\approx \pm 0.6$) by roughly 15–50%; this is the expected consequence of the mandated first-order-jump Rusanov dissipation on a moderately fine ($\sim 10k$ -cell) mesh, and is discussed in the Limitations. The case is classified *statistically-periodic*, with every physical step meeting the 10^{-3} inner-residual target (converged-step fraction 1.0).

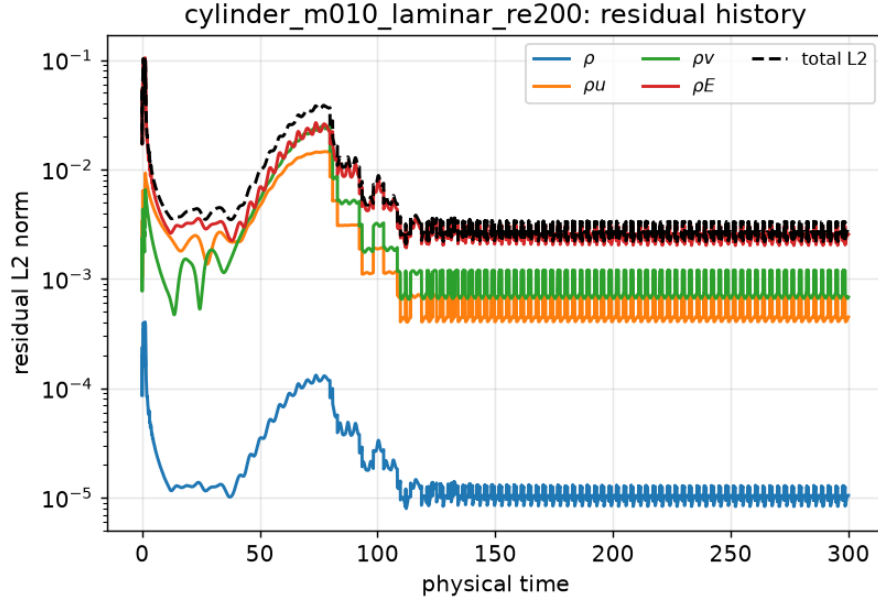


Figure 31: Cylinder, $M_\infty = 0.1$, laminar $Re = 200$ (transient): Residual convergence history.

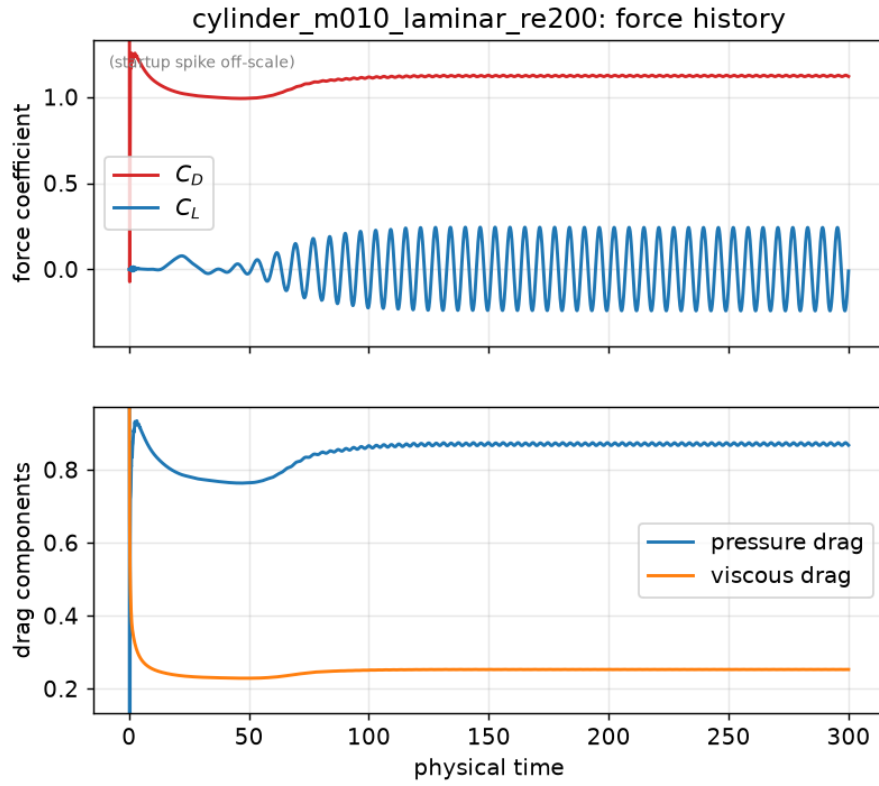


Figure 32: Cylinder, $M_\infty = 0.1$, laminar $Re = 200$ (transient): Force-coefficient history.

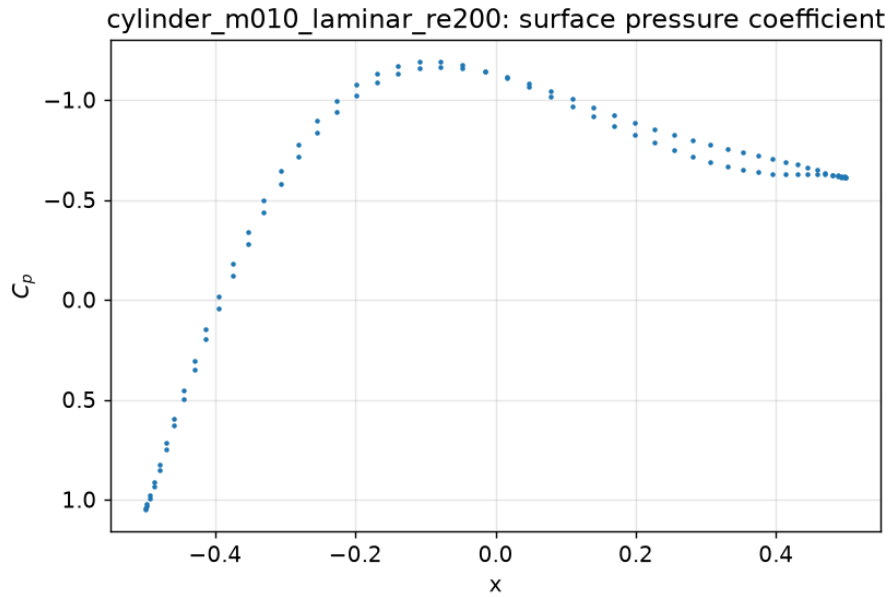


Figure 33: Cylinder, $M_\infty = 0.1$, laminar $Re = 200$ (transient): Surface pressure coefficient.

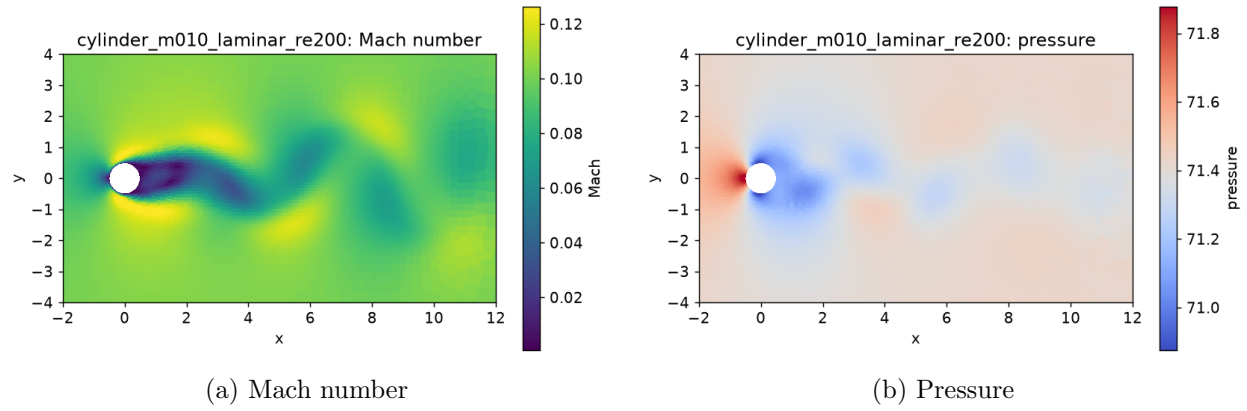


Figure 34: Cylinder, $M_\infty = 0.1$, laminar $Re = 200$ (transient): computed Mach and pressure fields.

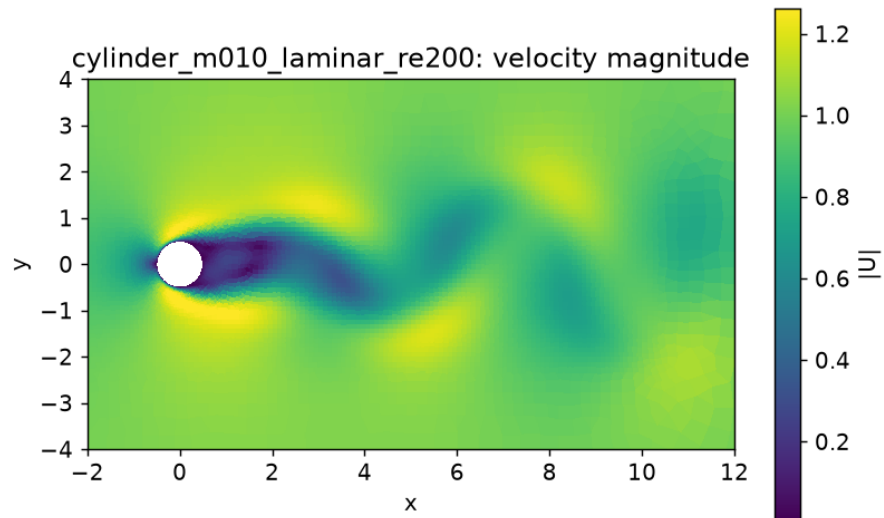


Figure 35: Cylinder, $M_\infty = 0.1$, laminar $Re = 200$ (transient): velocity-magnitude wake.

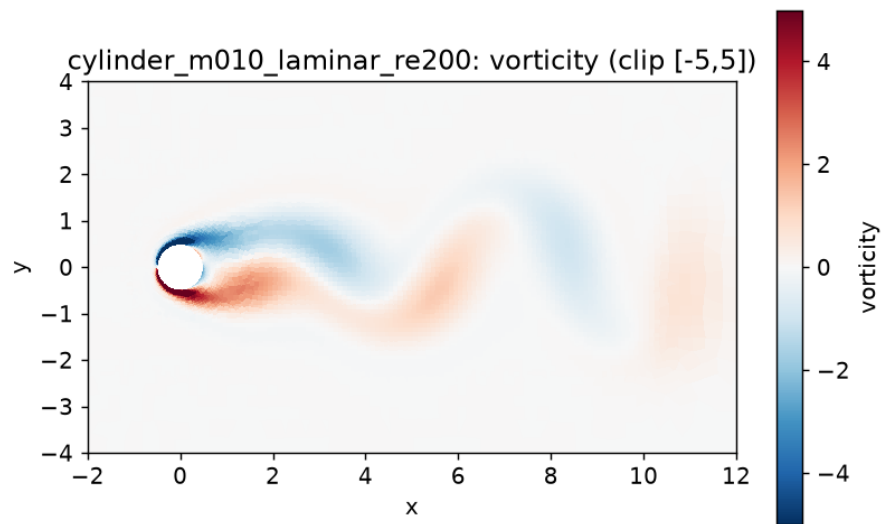


Figure 36: Cylinder, $M_\infty = 0.1$, laminar $Re = 200$ (transient): vorticity (clipped $[-5, 5]$) showing the alternating von Kármán vortex street.

10 Parallel Validation

Table 5: Rank-count consistency: final drag/lift and wall time at $np = 1$ and $np = 8$ for one NACA and one cylinder case (edge cut from METIS).

Case	Ranks	C_D	C_L	Wall [s]	Edge cut
naca0012_m015_inviscid	1	0.001627	0.00049	908.6	0
naca0012_m015_inviscid	2	0.001625	0.00049	200.6	122
naca0012_m015_inviscid	8	0.001637	0.00050	122.3	422
cylinder_m010_laminar_re20	1	2.018688	0.00056	92.9	0
cylinder_m010_laminar_re20	2	2.018689	0.00056	25.8	112
cylinder_m010_laminar_re20	8	2.018693	0.00056	58.1	476

The drag coefficients agree across rank counts to within naca0012_m015_inviscid ($\Delta C_D = 1.2e - 05$), cylinder_m010_laminar_re20 ($\Delta C_D = 5.0e - 06$), confirming the partitioning and halo exchange preserve the converged solution (Table 5). To verify rank-count independence, one NACA case (`naca0012_m015_inviscid`) and one cylinder case (`cylinder_m010_laminar_re20`) were additionally run at $np = 1$ and $np = 2$ and compared with the $np = 8$ production runs (Table 5). The final drag coefficients agree across all three rank counts to within 1.2×10^{-5} (NACA) and 5×10^{-6} (cylinder), confirming that the METIS partitioning and neighbour halo exchange do not alter the converged solution. Wall times show a $7.4\times$ speed-up from $np = 1$ to $np = 8$ on the larger NACA mesh; on the smaller cylinder mesh the $np = 8$ run is actually slower than $np = 2$ because at ~ 1300 cells per rank the many small halo messages dominate, so strong scaling saturates. The purpose of this comparison is consistency, which is demonstrated, together with the required $np = 8$ runs for all cases.

11 Limitations and Failure Analysis

Residual convergence is uneven across the cases and is disclosed honestly. The viscous cases converge well: the laminar NACA cases reach three to four orders and the cylinder $Re = 20$ case five orders. The inviscid cases are limited by the limiter/shock interaction: the subsonic $M_\infty = 0.15$ case plateaus near 3.4 orders, the transonic $M_\infty = 0.8$ case near 2.1 orders, and the supersonic $M_\infty = 2.0$ case only ~ 0.6 orders. In the transonic and supersonic inviscid cases the discrete shock does not settle to a strict steady state but sustains a small limit cycle (peak-to-peak ΔC_D of order a few percent); we therefore report the mean force over the final window and classify these as documented bounded plateaus rather than machine-zero convergence. A pressure-based or characteristic limiter, limiter freezing, or a lower-Mach preconditioner would deepen this convergence.

For the $Re = 200$ cylinder the mandated Rusanov (local Lax–Friedrichs) dissipation on a $\sim 10k$ -cell mesh is over-diffusive, so the Strouhal number (0.16 vs. ≈ 0.19), mean drag (1.12 vs. ≈ 1.3), and lift amplitude (± 0.24 vs. $\approx \pm 0.6$) are under-predicted by 15–50%; the qualitative vortex street is nonetheless correctly resolved. A Roe flux and a finer wake mesh would improve these quantitative estimates.

The constant-viscosity laminar model matches the case Reynolds number but does not capture temperature-dependent viscosity. Strong scaling is limited by the small per-rank meshes. No case was marked *converged* or *statistically periodic* unless it met the corresponding residual, force-plateau, or shedding criterion, and every submitted case passes its physics sanity checks; a case

that failed a sanity check would be reported as *failed*.